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9.3 Chords, Tangents, and Secants

Several clean theorems govern how chords, tangents, and secants interact with a circle. A radius drawn perpendicular to a chord always bisects it (and the converse: the perpendicular from the center to a chord bisects it), which lets the Pythagorean Theorem connect radius, chord, and distance-to-center. A tangent line is always perpendicular to the radius at the point of tangency, and the two tangent segments drawn from the same external point are always congruent. For intersecting chords, secants, and tangents, two families of theorems apply: **length** theorems (chord–chord: $a \cdot b = c \cdot d$; tangent–secant: $t^2 = (\text{whole})(\text{external})$; secant–secant: $(\text{whole}_1)(\text{external}_1) = (\text{whole}_2)(\text{external}_2)$ — together called the Power of a Point) and **angle** theorems (two chords meeting inside: angle $= \frac{1}{2}$(sum of intercepted arcs); two secants/tangents meeting outside: angle $= \frac{1}{2}$(difference of intercepted arcs)). This worksheet uses all of these, including multi-secant algebra and a real-world tangent application.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Radius 15, distance to chord 9. Find the chord length.

2. Tangent segments $4x + 1$ and 21. Find x.

3. Segments 9 and 8; other chord 6 and x. Find x.

4. Tangent 6; whole secant 18. Find the external part.

5. Intercepted arcs 120° and 40° outside the circle. Find the angle.

6. Radius 13, chord 10. Find the distance from center to chord.

7. Intersecting chords intercept arcs 80° and 50°. Find the angle.

8. Tangent segments $2x + 7$ and $x + 15$. Find x and the length.

9. Segments 4 and 15; other 6 and x. Find x.

10. Tangent 9; external part 3. Find the whole secant.

11. A radius meets a tangent at the point of contact. What angle is formed?

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12. Arcs 90° and 20° from two secants outside. Find the angle.

13. Radius 5, distance to chord 3. Find the chord length.

14. From external point P, one secant has near segment 5 and far segment x; another has near segment 6 and far segment 9. Find x.

15. An observer stands 13 km from the center of a circular pond of radius 5 km. Find the tangent line-of-sight distance to the pond's edge.

✓ Answer Key

1. 24
2. $x = 5$
3. $x = 12$
4. 2
5. 40°
6. 12
7. 65°
8. $x = 8$; length 23
9. $x = 10$
10. 27
11. 90°
12. 35°
13. 8
14. $x = 13$
15. 12 km